



PAYROLL RISK & EVIDENCE REVIEW

Cross-Module Integrity

Organisation	Organisation name not provided
Review period	01 Mar 2024 to 20 Apr 2024
Prepared as at	14 Apr 2026

CONFIDENTIAL

1. Executive Summary

This Cross-Module Integrity report focuses solely on inconsistencies identified between related payroll datasets, including employee lifecycle, leave activity, payroll events, and termination-related records.

A total of 8 cross-module integrity findings were identified across approximately 2 employees. These findings indicate possible linkage, sequencing, lifecycle, or dataset alignment weaknesses that may reduce confidence in the broader payroll data environment.

Across the dataset provided, the automated checks identified:

- **High:** 1
- **Medium:** 7
- **Low:** 0

A detailed breakdown by severity is provided in the **Findings Overview** section.

2. Data Sources

This review was generated from the following analysis outputs within the project `outputs/` directory:

- `cross_module_summary_by_severity.csv`
- `cross_module_findings.csv`
- `executive\executive_summary.md`
- `executive\executive_summary.json`

These outputs were produced by rule-based checks over payroll and HR CSV extracts supplied by the organisation for the review period.

3. Scope & Methodology

Modules included in this engagement:

- Cross-Module Integrity (CROSS_MODULE)
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Cross-Module Integrity – Scope & Methodology

Scope

The Cross-Module Integrity review assesses whether related payroll datasets align consistently across employee lifecycle, leave, payroll event, and termination records.

The purpose of this review is to identify inconsistencies between linked datasets that may indicate sequencing issues, lifecycle mismatches, incomplete integrations, or broader payroll data integrity weaknesses.

This review is designed to support payroll, HR, finance, and governance teams in identifying where records may not align cleanly across the broader payroll data environment. Findings are integrity signals only and do not, on their own, confirm non-compliance, underpayment, or payroll error.

Data reviewed

- employee master data (where supplied)
- leave balances and leave movement data (where supplied)
- payroll event / payroll transaction extracts (where supplied)
- termination and lifecycle-related records where included in the engagement data pack

Checks performed

- consistency checks between employee lifecycle status and payroll activity
- identification of mismatches between leave activity and termination or employment status
- cross-dataset linkage checks for related employee and payroll records
- detection of sequencing anomalies between linked events across modules

Out of scope

This review does not:

- calculate entitlements, underpayments or overpayments

- interpret awards, enterprise agreements, or employment contracts
- provide legal, accounting, or industrial relations advice
- assert contraventions of legislation or confirm non-compliance.

Cross-module findings should be interpreted as data integrity and linkage risk indicators. They highlight where records may not align cleanly across datasets and may require investigation before conclusions are drawn.

4. Findings Overview

Where a Cross-Module Integrity review was performed, the table below summarises the number of cross-module inconsistencies identified by severity. Counts reflect **integrity risk indicators only** and do not represent confirmed non-compliance or a quantified monetary impact.

Severity	Count	Description
HIGH	1	Cross-dataset inconsistencies that may materially affect confidence in employee lifecycle, payroll sequencing, or linked record integrity.
MEDIUM	7	Cross-module mismatches or data linkage issues that warrant review but may be explainable through timing, process, or source-system differences.
LOW	0	Lower-impact cross-module inconsistencies that should be monitored and improved over time.

5. Detailed Findings

This section sets out detailed findings for **Cross-Module Integrity** only. Findings highlight potential sequencing, lifecycle, and dataset-alignment issues across related payroll records. They are integrity indicators and do **not** on their own confirm non-compliance or incorrect pay outcomes.

CM-006

HIGH

Employee: E002 | Classification: LOGICAL | Dates: 2024-03-01 → 2024-04-10

FINDING

Payroll activity and leave ledger movement both continue after the employee termination date.

IMPACT

This may indicate data integrity, sequencing, or lifecycle mismatches across related payroll datasets. These issues can reduce confidence in linked records and make payroll outcomes or employee status changes harder to explain, validate, or reconcile.

RECOMMENDATION

Review the employee termination timeline, payroll activity, and leave transactions to confirm whether the employee was correctly finalised.

FINDING ID

1ba6ace90e24

EVIDENCE REFERENCE

```
{
  "sources": ["terminations.csv", "pay_events.csv", "leave_ledger.csv"],
  "primary_keys": {
    "employee_id": "E002",
    "termination_date": "2024-03-01"
  },
  "values": {
    "has_post_term_payroll_activity": true,
    "has_post_term_leave_activity": true,
    "latest_post_term_leave_date": "2024-04-10",
    "max_days_after_termination_pay": 50
  },
  "thresholds": {
    "allowed_days_after_term": 14,
    "high_severity_cutoff_days": 30
  },
  "explanation": "Payroll activity and leave ledger movement both continue significantly after termination, indicating a likely breakdown in offboarding control integrity."
}
```

CM-002

MEDIUM

Employee: E002 | Related record / leave type: ANNUAL | Classification: CONTEXTUAL | Dates: 2024-03-01 → 2024-04-10

FINDING

Leave ledger movement was recorded after the employee termination date.

IMPACT

This may indicate data integrity, sequencing, or lifecycle mismatches across related payroll datasets. These issues can reduce confidence in linked records and make payroll outcomes or employee status changes harder to explain, validate, or reconcile.

RECOMMENDATION

Review termination timing and leave processing history to confirm whether the movement is valid and properly evidenced.

FINDING ID

0b7e46d4abc1

EVIDENCE REFERENCE

```
{"sources": ["terminations.csv", "leave_ledger.csv"], "primary_keys": {"employee_id": "E002", "termination_date": "2024-03-01", "event_date": "2024-04-10", "leave_type": "ANNUAL"}, "values": {"event_type": null, "units": 8.0}, "explanation": "Leave ledger movement was recorded after the employee termination date."}
```

CM-002

MEDIUM

Employee: E005 | Related record / leave type: ANNUAL | Classification: CONTEXTUAL | Dates: 2024-03-01 → 2024-03-15

FINDING

Leave ledger movement was recorded after the employee termination date.

IMPACT

This may indicate data integrity, sequencing, or lifecycle mismatches across related payroll datasets. These issues can reduce confidence in linked records and make payroll outcomes or employee status changes harder to explain, validate, or reconcile.

RECOMMENDATION

Review termination timing and leave processing history to confirm whether the movement is valid and properly evidenced.

FINDING ID

3f3de4c61b56

EVIDENCE REFERENCE

```
{"sources": ["terminations.csv", "leave_ledger.csv"], "primary_keys": {"employee_id": "E005", "termination_date": "2024-03-01", "event_date": "2024-03-15", "leave_type": "ANNUAL"}, "values": {"event_type": null, "units": 4.0}, "explanation": "Leave ledger movement was recorded after the employee termination date."}
```


CM-012

MEDIUM

Employee: E002 | Related record / leave type: ANNUAL | Classification: CONTEXTUAL | Dates: 2024-03-01 → 2024-04-10

FINDING

Leave ledger activity was recorded after the employee termination date.

IMPACT

This may indicate data integrity, sequencing, or lifecycle mismatches across related payroll datasets. These issues can reduce confidence in linked records and make payroll outcomes or employee status changes harder to explain, validate, or reconcile.

RECOMMENDATION

Confirm whether the employee was reinstated or whether the ledger events were posted incorrectly.

FINDING ID

ca37a32075d0

EVIDENCE REFERENCE

```
{"sources": ["leave_ledger.csv", "terminations.csv"], "primary_keys": {"employee_id": "E002", "termination_date": "2024-03-01", "event_date": "2024-04-10", "leave_type": "ANNUAL"}, "values": {"event_type": "nan", "units": 8.0, "days_after_termination": 40}, "thresholds": {"allowed_event_types": ["ADJUSTMENT", "PAYOUT"]}, "explanation": "Leave ledger activity was recorded after the employee termination date."}
```

CM-012

MEDIUM

Employee: E005 | Related record / leave type: ANNUAL | Classification: CONTEXTUAL | Dates: 2024-03-01 → 2024-03-15

FINDING

Leave ledger activity was recorded after the employee termination date.

IMPACT

This may indicate data integrity, sequencing, or lifecycle mismatches across related payroll datasets. These issues can reduce confidence in linked records and make payroll outcomes or employee status changes harder to explain, validate, or reconcile.

RECOMMENDATION

Confirm whether the employee was reinstated or whether the ledger events were posted incorrectly.

FINDING ID

d73511e58fab

EVIDENCE REFERENCE

```
{"sources": ["leave_ledger.csv", "terminations.csv"], "primary_keys": {"employee_id": "E005", "termination_date": "2024-03-01", "event_date": "2024-03-15", "leave_type": "ANNUAL"}, "values": {"event_type": "nan", "units": 4.0, "days_after_termination": 14}, "thresholds": {"allowed_event_types": ["ADJUSTMENT", "PAYOUT"]}, "explanation": "Leave ledger activity was recorded after the employee termination date."}
```

CM-016

MEDIUM

Employee: E002 | Classification: STRUCTURAL | Termination: 2024-03-01

FINDING

A termination record exists but no supporting leave snapshot record was identified for the employee.

IMPACT

This may indicate data integrity, sequencing, or lifecycle mismatches across related payroll datasets. These issues can reduce confidence in linked records and make payroll outcomes or employee status changes harder to explain, validate, or reconcile.

RECOMMENDATION

Review snapshot extract completeness and confirm whether terminated employees were intentionally excluded or whether leave balances were missing from the extract.

FINDING ID

ca93e3ef4ed9

EVIDENCE REFERENCE

```
{"sources": ["terminations.csv", "balances_snapshot.csv"], "primary_keys": {"employee_id": "E002", "termination_date": "2024-03-01"}, "values": {"leave_snapshot_record_found": false}, "thresholds": {"leave_types_checked": ["ANNUAL", "LSL"]}, "explanation": "A termination record exists but no supporting leave snapshot record was identified for the employee."}
```

CM-016

MEDIUM

Employee: E005 | Classification: STRUCTURAL | Termination: 2024-03-01

FINDING

A termination record exists but no supporting leave snapshot record was identified for the employee.

IMPACT

This may indicate data integrity, sequencing, or lifecycle mismatches across related payroll datasets. These issues can reduce confidence in linked records and make payroll outcomes or employee status changes harder to explain, validate, or reconcile.

RECOMMENDATION

Review snapshot extract completeness and confirm whether terminated employees were intentionally excluded or whether leave balances were missing from the extract.

FINDING ID

81c9d3c56fec

EVIDENCE REFERENCE

```
{"sources": ["terminations.csv", "balances_snapshot.csv"], "primary_keys": {"employee_id": "E005", "termination_date": "2024-03-01"}, "values": {"leave_snapshot_record_found": false}, "thresholds": {"leave_types_checked": ["ANNUAL", "LSL"]}, "explanation": "A termination record exists but no supporting leave snapshot record was identified for the employee."}
```

CM-020**MEDIUM**

Employee: E002 | Classification: CONTEXTUAL

FINDING

An employee triggered multiple cross-module integrity failures.

IMPACT

This may indicate data integrity, sequencing, or lifecycle mismatches across related payroll datasets. These issues can reduce confidence in linked records and make payroll outcomes or employee status changes harder to explain, validate, or reconcile.

RECOMMENDATION

Review the full lifecycle of the employee including payroll, leave, termination and evidence records.

FINDING ID

050e64ad6d2a

EVIDENCE REFERENCE

```
{"total_findings": 4, "high_findings": 1, "thresholds": {"min_findings": 3, "min_high_severity": 2}, "explanation": "The employee triggered multiple cross-module integrity failures, indicating a clustered lifecycle control issue that should be reviewed in context."}
```

6. Limitations & Assumptions

This review is subject to the following limitations:

- Calculations assume the underlying pay rates, loadings and multipliers are correct in the source systems.
- Award and enterprise agreement interpretation is not performed by this tool.
- Holiday calendars, leave rules and accrual settings are assumed to reflect the organisation's intended configuration.
- Data quality issues (missing records, duplicates, inconsistent identifiers) may affect the completeness and accuracy of the results.

7. Recommended Next Steps

1. Validate the highest-severity findings first.
 2. Review the most affected modules and confirm whether findings reflect genuine control issues or data limitations.
 3. Address structural data gaps that weaken evidentiary confidence.
 4. Confirm root causes before remediation.
 5. Re-run the review after corrective action to confirm that risk indicators have reduced.
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8. Appendices

Appendix A – Rule Definitions

This review used a set of automated rules to flag evidential and process risk indicators.

Cross-Module Integrity (CROSS_MODULE)

- Employee lifecycle mismatches across datasets
 - Leave activity inconsistent with employment or termination status
 - Payroll events inconsistent with linked employee or termination records
 - Cross-dataset linkage or sequencing anomalies
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Appendix B – Machine-readable outputs

Complete machine-readable outputs are available in the generated CSV and summary files for the modules included in this engagement.

These files provide row-level detail suitable for operational review, sampling, remediation planning, or incorporation into a broader audit work program.
